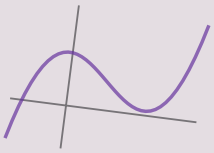
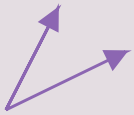
PREVIEW

$y = Ax^n$

$u_n = a + (n-1)d$

$d^2y/dx^2 < 0$

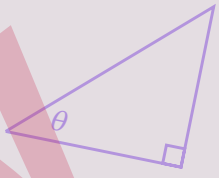






$A = \frac{1}{2}r^2\theta$





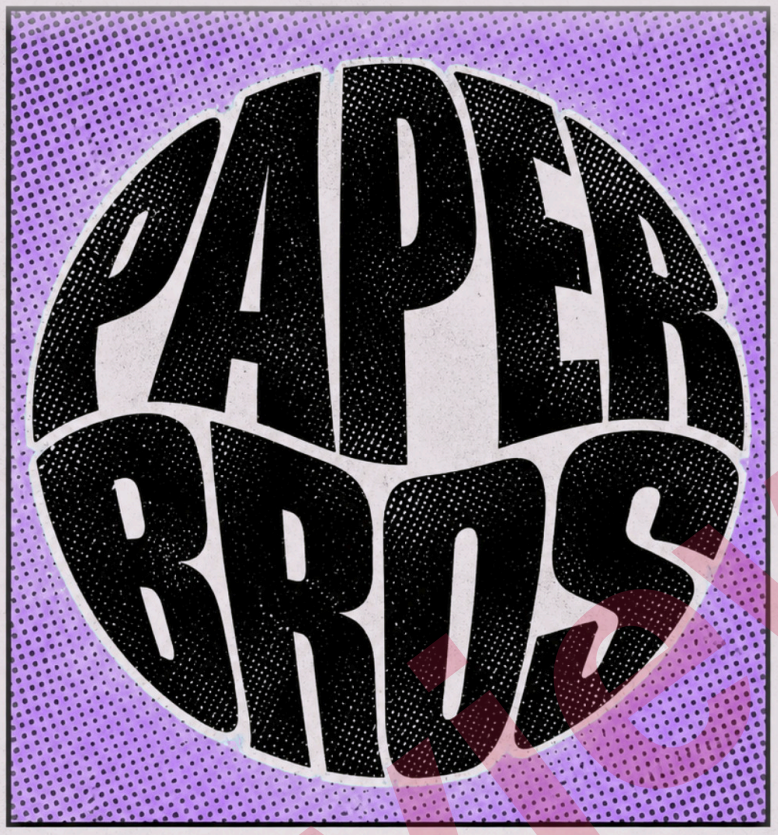
$v = ds/dt$

$(a+b)^n$

sec^2x



$\ln x$



A ONE STOP SOLUTION FOR YOUR BOARDS

DEVELOPED BY STUDENTS, FOR STUDENTS

CAMBRIDGE O LEVEL

ADDITIONAL MATHEMATICS

4037 · PAPER 2

TOPICAL QUESTIONS & MARK SCHEMES

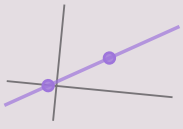
MAY/JUNE 2015 – MAY/JUNE 2026

530 QUESTIONS · FULL WORKING SPACE · MARK SCHEMES FROM PAGE 677

Preview

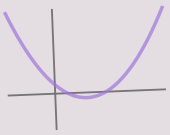
$\sin^2\theta + \cos^2\theta = 1$

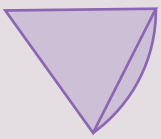




e^x

$s = r\theta$







^nC_r

$S\infty = a/(1-r)$

$f^{-1}(x)$



PREVIEW

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Additional Mathematics 4037 · Paper 2 · Questions

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Preview

PREVIEW

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Preview

PREVIEW

Topic 1: Functions

1.1 Function terminology

0 entries

No questions on this subtopic in these papers.

1.2 Domain and range

5 entries

Q1 **2017/May-June/22/Q12 (ii)(v)**

earlier parts included for context (marked with a bar)

12 The function g is defined, for $x > -\frac{1}{2}$, by $\mathrm{g}(x) = \frac{3}{2x+1}$.

(i) Show that $\mathrm{g}'(x)$ is always negative. [2]

(ii) Write down the range of g. [1]

The function h is defined, for all real x, by $\mathrm{h}(x) = kx + 3$, where k is a constant.

(iii) Find an expression for $\mathrm{hg}(x)$. [1]

(iv) Given that $\mathrm{hg}(0) = 5$, find the value of k. [2]

(v) State the domain of hg. [1]

Preview

PREVIEW

Q2 **2018/May-June/21/Q5 (ii)** earlier parts included for context (marked with a bar)

5 The function f is defined by $f(x) = \frac{1}{2x-5}$ for $x > 2.5$.

(i) Find an expression for $f^{-1}(x)$. [2]

(ii) State the domain of $f^{-1}(x)$. [1]

Q3 **2021/May-June/24/Q9 (a)**

9 A function f is defined, for all real values of x, by $f(x) = 3 + e^{5x}$.

(a) Find the range of f. [1]

Q4 **2025/Oct-Nov/22/Q9 (a)(b)**

9 It is given that $f(x) = \ln(2x+5)$ for $x > a$, where a is a constant.

(a) Write down the least possible value of a. [1]

(b) Using your value of a, write down the range of f. [1]

Preview